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PHYS-3701 Electronics-I B.S. 4 year Spring 2023

Short Questions

(i) If the dc current gain of a transistor is 150, find the β_{dc} and α_{DC}.

dc current gain = 150

$\beta_{dc} = ?$

$\alpha = ?$

$\beta_{dc} =$ dc current gain $= 150$

$$\alpha_{dc} = \frac{\beta_{dc}}{\beta_{dc}+1} = \frac{150}{150+1} = \frac{150}{151}$$

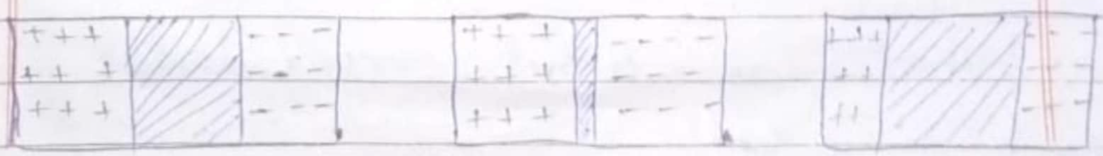
$$\alpha_{dc} \approx 0.99338.$$

(ii) Compare the depletion region in forward & reverse bias.

In forward bias condition, the width of depletion region is decreased because ~~majority~~ is movement of both electrons and holes inside the crystal, only free electrons move in external circuit. Due to applied external voltage, the barrier potential is reduced which allows more current to flow across the junction. Therefore forward bias reduces the thickness of depletion region.

While in reverse bias condition, width of depletion region is increased because majority charge carriers are pulled away from the junction.

No bias Forward bias Reverse bias



(2)

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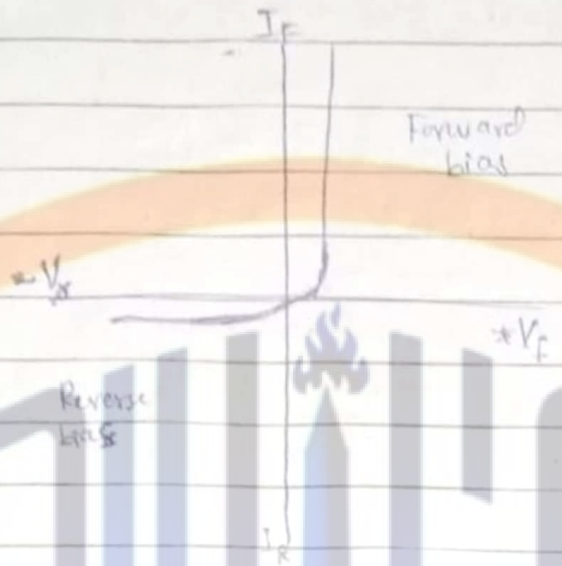
(iii) Draw characteristics curves of practical diode model.



(iv) How does full wave voltage differ from a half wave voltage?

	Full wave	Half wave
1.	Full wave voltage occurs on each half of the input.	A half wave voltage occurs once each input cycle.
2.	A full wave rectifier has a frequency twice the input frequency.	A half wave rectifier has a frequency equal to input frequency.

(v) Describe briefly under what conditions do we use a π-filter?

We use π-filter circuit when:

(i) We want to use lower cost filter circuit
(ii) Overload current is small.

(vi) What is a hot carrier diode?

A hot carrier diode, also, known as

a schottky diode, is a semiconductor diode that is made by joining a metal and a semiconductor. It has a metal semiconductor junction instead of p-n junction, and the interface between the two is called a schottky barrier or schottky contact.

(vii) **Define Q-point?**

When there is no signal, then the point on the load line at which the transistor is working is called a Q-point, also known as quiescent or operating point.

(viii) **What are the two advantages of voltage divider bias?**

The advantages of voltage divider bias are

1. It is stable.
2. It require only one supply voltage.

(ix) **What value must gate to source voltage ~~does~~ have to produce cut-off in a p-channel JFET with a pinch off voltage -3V.**

To produce cut off in a p-channel JFET with a pinch off voltage of -3V, the gate to source voltage (V_{GS}) must be equal to the pinch off voltage.

$$V_{GS} = V_p = -3V.$$

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(x) Between what two voltages does a current regulator diode operate?

A current regulating diode also known as a current limiting diode or constant current diode operates between a voltage range of less than 1V to 100V.

(xi) When a transistor is used as a switch, in what two states is it operated?

When a transistor is used as a switch, the two states in which it operates are cut off and saturation.

(xii) If the gate to source voltage in an n-channel E-MOSFET is made more positive, does the drain current increase or decrease?

When the gate-to-source voltage in an n-channel E-MOSFET is made more positive, then the drain current increases.

(xiii) How is voltage gain defined?

Voltage gain is the ratio of output voltage to the input voltage in an amplifier circuit.

$$A_V = \frac{V_o}{V_i}$$

A_V = The gain

V_o = The output voltage

V_i = The input voltage.

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(xiv) One FET has a transconductance of 3000 μS and other has a transconductance of 3.5 mS. Which one can produce the higher voltage gain, with all other circuit components the same?

In FET, the voltage gain is determined by

$$A_v = -g_m R_L$$

g_m = transconductance, R_L = load resistance

As all other components are same for both FETs, the voltage gain will be directly proportional to transconductance. So the FET with higher transconductance will produce the high voltage gain.

FET 1 has $g_m = 3000 \times 10^{-6}\,S = 3\,mS$

FET 2 has $g_m = 3.5 \times 10^{-3}\,S = 3.5\,mS$

The FET 2 will produce higher voltage gain than FET

(xv) What are the applications of clampers?

1. Clampers can be used as voltage doublers or voltage multipliers.
2. Clampers can be used for removing the distortions.
3. Clampers are frequently used in test equipment, sonar and radar system.
4. For the protection of the amplifiers from large errant signals clampers are used.
5. For improving the overdrive recovery time clampers are used.

Long Questions:-

Q#1 (a).

What are clippers? Discuss the operation

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of positive and negative clippers in detail.
Give at least one example of clipper. (5)

From ilmi book page # 34

(b).

How can we use zener diode as voltage regulator? (5)

From Ilmi book page #12 /

From MBD page #37

Q#2 (a).

Briefly explain two types of MOSFET. (5)

From ilmi book page # 130 /

From MBD page # 97

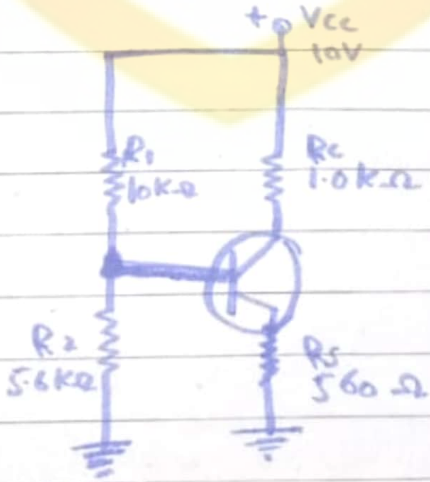
(b).

Name different biasing methods of MOSFET. Analyze one of them. (5)

From Ilmi chap#6

Q#3 (a)

Determine I_c and V_{CE} in the circuit of figure 1. The transistor has a $\beta_{DC} = 150$.



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The base voltage is

$$V_B \cong \left(\frac{R_2}{R_1+R_2}\right) V_{cc}$$

$$V_B = \left(\frac{5.6\,K\Omega}{15.6\,K\Omega}\right) 10V$$

$$V_B = 3.59V$$

So,

$$V_E = V_B - V_{BE}$$

$$V_E = 3.59V - 0.7V$$

$$V_E = 2.89V$$

and

$$I_E = \frac{V_E}{R_E}$$

$$I_E = \frac{2.89V}{560\Omega}$$

$$I_E = 5.16mA$$

Therefore,

$$\boxed{I_c \simeq I_E = 5.16mA}$$

and

$$V_c = V_{cc} - I_c R_c$$

$$V_{cc} = 10V - (5.16mA)(1.0K\Omega)$$

$$V_{cc} = 4.84V$$

$$V_{CE} = V_{cc} - V_E$$

$$V_{CE} = 4.84V - 2.89V$$

$$\boxed{V_{CE} = 1.95V}$$

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(b)

What is the phase relationship of the input and output voltages of a common emitter amplifier. (2)

The phase difference of 180° between the signal voltage and output voltage in a common-emitter amplifier is phase reversal. The signal is fed at the input terminals base and emitter and output is taken from collector and emitter in common emitter configuration of amplifier. The amplification is not affected by this phase reversal.

No phase reversal of voltage occurs in common-base and common-collector amplifier. The ac output voltage is in phase with an ac input signal. For all three amplifier configurations, input and output currents are in phase.